

COMP 5533 DATA MINING — ASSIGNMENT 2

Put a picture/screenshot that best illustrates your work (e.g., dataset, model diagram, or workflow).

Std-No. — Name Surname: _____

Submission Date: _____

Focus: Decision Trees, Bayesian Classifiers & Networks, Support Vector Machines (SVM), and Clustering

Format: Individual; all writing/code/figures must be your own or fully understood and properly cited.

Submission:

- (1) 10–20-page PDF/docx report
- (2) Reproducible code.
- (3) 5 slides or a 5-minute video with key insights and at least one failure case.

Assignment Learning Objectives

- ✓ Explain/derive nontrivial results about decision trees, Bayesian classifiers/networks, SVMs, and clustering (consistency, calibration, margins, identifiability).
- ✓ Design rigorous experiments: robustness to shift/noise/outliers, and model selection for both supervised and unsupervised settings.
- ✓ Bridge theory and practice; critically engage with classic and recent literature.

Abstract

Briefly explain what you do and why it matters, and what you learned (≥ 50 –100 words). Considering the dataset(s), models, methods, and the key research question(s) from the literature.

Part A — Theory (rigorous, self-contained explanation/derivations)

1. Impurity axioms: Prove that Gini and entropy satisfy symmetry, maximality at uniform class distribution, and refinement (splitting reduces expected impurity). Provide a counterexample for a non-admissible impurity.
2. Bayes-consistency: Discuss conditions under which greedily-grown trees with appropriate pruning depths are Bayes-consistent. Provide a toy distribution where unlimited-depth trees overfit while a properly pruned tree approaches Bayes risk.
3. Given a decision tree, you have the option of (a) converting the decision tree to rules and then pruning the resulting rules, or (b) pruning the decision tree and then converting the pruned tree to rules. What advantage does (a) have over (b)?
4. Why is naïve Bayesian classification called “naïve”? Briefly outline the major ideas of naïve Bayesian classification.
5. Naive Bayes optimality: Under class-conditional feature independence, derive the Maximum A Posteriori (MAP) classifier. Construct a dataset where Naive Bayes is Bayes-optimal, and another where dependence breaks optimality but NB still generalizes well in finite samples (bias-variance argument).
6. k-means: show that Lloyd’s algorithm monotonically decreases the k-means objective and can converge to local minima; provide a concrete dataset where it reaches a suboptimal solution.
7. DBSCAN: Define core, border, and noise points. show that with fixed ϵ and minPts, the output is invariant to permutation of the input and discuss conditions under which two clusters will be merged as ϵ increases (connectivity percolation argument).

Part B — Practice (empirical study)

Use one tabular dataset and one text or image dataset (moderate size).

Examples: UCI Adult/Credit/Breast Cancer; IMDB/AG News (bag-of-words or embeddings); MNIST/Fashion-MNIST.

8. **Implement and compare:** (1) Decision Tree (CART); (2) Naive Bayes (Gaussian or Multinomial) and one simple Bayesian Network (e.g., Tree-Augmented Naive Bayes or a small hand-crafted DAG); (3) SVM with linear and RBF kernels (with scaling).
9. **Compute metrics:** such as Accuracy, Precision/Recall/F1, ROC-AUC or PR-AUC (state class balance), and Expected Calibration Error (ECE). Include confusion matrices and reliability curves.
10. **Compare** at least three clustering methods: k-means, GMM-EM, spectral clustering (justify kernel scale), plus one of DBSCAN/HDBSCAN or agglomerative (Ward).

Part C — Literature Engagement

Pick one paper per family (decision trees, Bayesian, SVM, clustering). For each:

- (i) 1–2 page critical summary;
- (ii) replicate a small empirical claim or provide a counterexample;
- (iii) propose one extension and justify.

Grading Rubric (100 points)

Component	Points
A. Theory	30 (correctness & depth 18, clarity 6, insightful counterexamples 6)
B. Practice	40 (design/baselines 10, analysis 18, reproducibility 8, failure analysis 4)
C. Literature	30 (critique 8, replication/counterexample 8, extension 4, 10 (coherence, originality, clarity))

Academic Integrity

- You may read and cite any source; write in your own words and cite clearly.
- Code must be your own unless explicitly credited; include a contributions paragraph (including any LLM assistance).